Prabhavit Gupta



Cloud & AI Engineer: Python APIs, RAG Architectures & LLM Systems

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Summary

Cloud and AI Engineer with 3 years of production software engineering experience architecting scalable Python APIs, Retrieval-Augmented Generation (RAG) architectures, and cloud-integrated LLM systems. Hands-on expertise developing production frameworks with FastAPI, LangChain, vector retrieval pipelines, and asynchronous messaging systems. Resident in Venice, Italy with an EU Long-Term Residence Permit (permanent unrestricted Italian and EU work authorization, no sponsorship required). Bilingual English (C1 Professional), Italian (conversational/parlato); available immediately for full-time cloud and AI engineering roles.

Technical Expertise

- **AI Engineering & LLM Systems:** Architetture RAG, LangChain, OpenAI / Claude API integration, vector embeddings, semantic search, hybrid retrieval, prompt engineering, agentic function calling.
- **Cloud Infrastructure & Orchestration:** Cloud engineering, Docker containerization, asynchronous messaging (Pub/Sub, Redis queues), RESTful API design, CI/CD pipeline automation, GitOps.
- **Backend Development & Python:** Python (FastAPI, asyncio, pandas, NumPy, Pydantic), TypeScript, Node.js, microservices architecture, JSON schema enforcement, database modeling.
- **Databases & Data Management:** PostgreSQL (relational modeling, indexing, query optimization), Redis caching, vector stores, data extraction and ETL transformation pipelines.
- **Testing & Observability:** End-to-end integration testing with Playwright, API contract verification, cloud monitoring, structured logging, latency and cost optimization.
- **Languages & Authorization:** Italian (Parlato/Conversational), English (C1 Professional, fluent), Hindi (Madrelingua). Permesso di Soggiorno UE a tempo indeterminato (autorizzazione illimitata al lavoro in Italia e UE).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Engineered production cloud AI services and RAG search pipelines for a marketplace platform serving 4,500+ active enterprise users across Europe.
- Developed high-throughput Python (FastAPI) and Node.js backend APIs, integrating vector retrieval and LLMs with internal PostgreSQL databases.
- Built LangChain orchestration workflows with automated context injection and tool execution, reducing edge-case query failure rates.
- Implemented semantic caching and asynchronous task processing using Redis and message queues, optimizing cloud resource consumption and query latency.
- Deployed Docker containers across cloud infrastructure with automated CI/CD deployment pipelines and comprehensive API contract testing.
- Managed monitoring, distributed logging, and incident response for AI API microservices, maintaining high operational availability.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and delivery of cloud-integrated web services and data processing pipelines for European clients.
- Built automated ETL data ingestion services connecting external APIs to relational databases, ensuring consistent data synchronization.
- Developed interactive prototypes to validate API specifications and architecture designs prior to production deployment.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Automation & Integration Developer *Venice, Italy*

- Designed and deployed custom Python automation microservices, webhook listeners, and LLM integrations for SMB clients in Italy and Europe.

- Automated document processing and structured data extraction workflows, replacing fragile manual data entry with resilient event-driven systems.
- Authored technical architecture documentation, API guides, and system configuration manuals for client engineering teams.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.